A NONZERODIVISOR AVOIDING THE MINIMAL PRIMES OVER THE MAXIMAL IDEAL HITS THE GENERIC POINT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 14717-14752, immediately following the statement at 14701-14715.

Editorial note. The proof is by contradiction and ends with the explicit contradiction against hypothesis (3); that final step is inside the span.

Editorial note. Three cross-chapter pointers (Algebra, Morphisms) plus a same-chapter lemma are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Locally principal vertical parts. chapter lemma lemma-locally-principal-vertical, cited at proof lines 14744-14745. Applied to Spec(C) -> Spec(A) and Z = Spec(C/gC) to produce the contradiction that closes the proof.
- Image of a finite map and generic points. Algebra, lemma algebra-lemma-image-dense-generic-points, cited in the proof. Controls the generic point of Spec(A) as a point of the image of the restricted morphism.
- Dimension of the fibre at a point. Morphisms, lemma morphisms-lemma-dimension-fibre-at-a-point, cited in the proof. Supplies the fibre-dimension interpretation of the hypothesis (4) on dimensions and transcendence degrees.

Source reference. Original reference 00FL: focused-seeds/geometry/S9-morph-horizontal/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 053R: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14538–14538. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02FX: focused-seeds/geometry/S9-morph-horizontal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $A \to B$ be a local homomorphism of local rings, and $g \in \mathfrak{m}_B$. Assume*

(1) *A and B are domains and $A \subset B$,*
(2) *B is essentially of finite type over A,*
(3) *g is not contained in any minimal prime over $\mathfrak{m}_A B$, and*
(4) *$\dim(B/\mathfrak{m}_A B) + trdeg_{\kappa(\mathfrak{m}_A)}(\kappa(\mathfrak{m}_B)) = trdeg_A(B)$.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Then $A \subset B/gB$, i.e., the generic point of $\mathrm{Spec}(A)$ *is in the image of the morphism* $\mathrm{Spec}(B/gB) \to \mathrm{Spec}(A)$.

Proof. Note that the two assertions are equivalent by Algebra, Lemma 00FL. To start the proof let C be an A-algebra of finite type and $\mathfrak{q}$ a prime of C such that $B = C_{\mathfrak{q}}$. Of course we may assume that C is a domain and that $g \in C$. After replacing C by a localization we see that $\dim(C/\mathfrak{m}_A C) = \dim(B/\mathfrak{m}_A B) + \mathrm{trdeg}_{\kappa(\mathfrak{m}_A)}(\kappa(\mathfrak{m}_B))$, see Morphisms, Lemma 02FX. Setting K equal to the fraction field of A we see by the same reference that $\dim(C \otimes_A K) = \mathrm{trdeg}_A(B)$. Hence assumption (4) means that the generic and closed fibres of the morphism $\mathrm{Spec}(C) \to \mathrm{Spec}(A)$ have the same dimension.

Suppose that the lemma is false. Then $(B/gB) \otimes_A K = 0$. This means that $g \otimes 1$ is invertible in $B \otimes_A K = C_{\mathfrak{q}} \otimes_A K$. As $C_{\mathfrak{q}}$ is a limit of principal localizations we conclude that $g \otimes 1$ is invertible in $C_h \otimes_A K$ for some $h \in C$, $h \notin \mathfrak{q}$. Thus after replacing C by C_h we may assume that $(C/gC) \otimes_A K = 0$. We do one more replacement of C to make sure that the minimal primes of $C/\mathfrak{m}_A C$ correspond one-to-one with the minimal primes of $B/\mathfrak{m}_A B$. At this point we apply Lemma 053R to $X = \mathrm{Spec}(C) \to \mathrm{Spec}(A) = S$ and the locally closed subscheme $Z = \mathrm{Spec}(C/gC)$. Since $Z_K = \emptyset$ we see that $Z \otimes \kappa(\mathfrak{m}_A)$ has to contain an irreducible component of $X \otimes \kappa(\mathfrak{m}_A) = \mathrm{Spec}(C/\mathfrak{m}_A C)$. But this contradicts the assumption that g is not contained in any prime minimal over $\mathfrak{m}_A B$. The lemma follows. $\square$